Engineering Mathematics - I · Unit_03 · Unit 3

DRAFT — UE26MA141A · Unit_03 · Unit 3 · Particular integral of type 1 and 2

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Particular integral of type 1 and 2

1 Particular integral of type 1 and 2

Study focus: Particular integral of type 1 and 2. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000138 · EV0000139

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we will learn.

Key details

- And f of x comma y equals sine of a x plus b y as well as cos of a x plus b y.
- It means this is not zero.
- That is z equals 1 by f of d comma d dash.
- Then f of x comma y.
- So basically what we do is that we need to take up a and b from here and then replace b by a and d dash by b so the particular integral becomes one over f of a comma b tim a power a x plus b y remember we have a condition here f of a comma b.
- Should be non-zero.
- Otherwise that leads to one by zero case.
- That is Z is written as.
- Some foyer of x comma y times e power a plus b y.
- Where this faya of x comma y is nothing but f of b plus a and d dash plus b.
- It means we have d and d dash.
- Once we write foyer of x comma y to be f of b plus a d dash plus b we need to expand this as an infinite series.
- We substitute back to Z equals phi of x comma y times e power a x plus b y.
- D inverse is nothing but integral with respect to x, keeping y constant, and d dash inverse is nothing but integral with respect to y, keeping x constant.
- Find the general solution for the partial differential equation 2d square minus dd dash minus d dash square plus d.

Example or application from the lecture

- How to find the particular integral for the given partial differential equation where the right hand side is non-zero.
- As we know that the general form of the non-homogeneous partial differential equation is given by f of b, d dash of z equals f of x comma y.
- In this case, the formula for particular integral is given by z equals 1 over f of d comma d dash times f of x gamma y.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000138 · EV0000139 · EV0000140 · +1 source units in pack.json

Core explanation · 05:00–09:59

Minus D dash equals B power two x plus three y.

Key details

- It is easy to see that a values two and b values three.
- differently partially different equation can be written as 2D plus 1 and D minus D dash.
- So for the first factor, we can see this is A1 value.
- B1 is 1 and C1 is 1.
- For the second factor, A2 is 1 b2 is minus 1 and c2 is 0.
- And we know that in the divine problem a values two and b value is three, when you substitute b s two and d dash to be three.
- [ASR low confidence; check audio] e power 2x plus 3y.
- 4 d square plus 3 d dash minus d dash square minus d minus d dash of z equals this.
- As you can see a value is one by two and b values two.
- And it is easy to see that.
- When we replace d by half and d dash by two, so we get zero.
- That is phi of x comma y.
- and e power x plus two y by two.
- When you substitute this back to.

Example or application from the lecture

- Let us find out the particular integral.
- In the given problem that is f of d comma d dash, we see that this is going to be minus 8, so which is non-zero.
- And hence the particular integral for this particular case is given by minus 1 by 8 times e power 2x plus 3 by.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000142 · EV0000143 · EV0000144 · +1 source units in pack.json

Core explanation · 10:00–14:59

And we know from previous slide.

Key details

- F comma d dash to contain d square, d dash and d dash square.
- D dash by minus a b and a dash square by minus b square.
- So the same method can be used even for cost of a x plus b y.
- In case we get denominator zero, that should be solved by general case, which we will see later.
- That is general case.
- 2 times dough Z by doux square minus 3 dou square z by doux dou y plus dou square z by dou y square.
- And the very well this is same as d square and this is same as the d dash and this is sam as the dash square.
- Are the total degree same.
- We see that.
- When we replace d square by minus one square, we get minus one.
- D dash by exactly minus two, so that become plus two here, then d dash square to be minus of minus two square, so that becomes totally minus four.
- Is minus one by twelve sine of x minus two y.

Example or application from the lecture

- Then we take out a and b from the given problem, then replace d square by minus a square.
- So let's take our example.
- Find the particular integral for the given partial differential equation.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000146 · EV0000147 · EV0000148 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Particular integral of type 1 and 2 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Particular_integral_of_type_1_and_2__265cba67cfcb2e18.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Particular integral of type 1 and 2 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Particular_integral_of_type_1_and_2__265cba67cfcb2e18.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral of type 1 and 2 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Particular_integral_of_type_1_and_2__265cba67cfcb2e18.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9981; only the JSON segmentation supplies the displayed timestamps.
Particular integral of type 1 and 2 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Particular_integral_of_type_1_and_2__265cba67cfcb2e18.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00042	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0054. Sidecar variants compare...
SRC00043	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0054. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00044	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0054. Sidecar variants compare...
SRC00045	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0054. Sidecar variants compare...

C Revision and retrieval practice